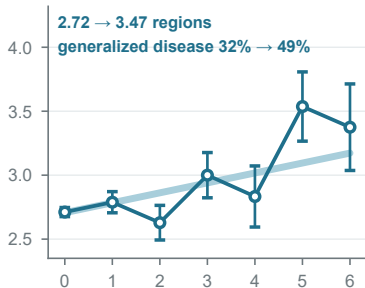


A**Genes that act widely cause disease widely**

Body regions affected by the Mendelian disease (of 6)

2.72 → 3.47 regions

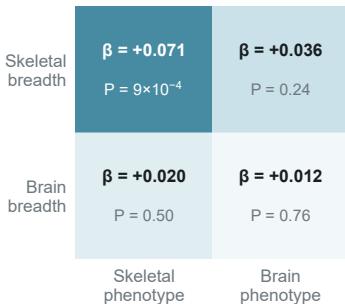
generalized disease 32% → 49%



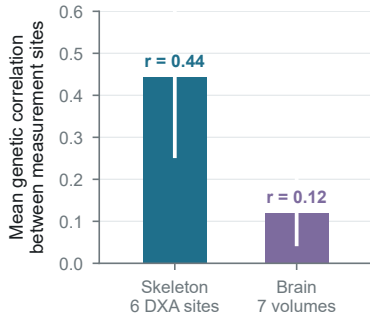
Breadth of action (number of skeletal sites)

n = 2,789 genes $\beta = 0.078$ per site $P = 3 \times 10^{-4}$ research intensity ruled out: $\rho(\text{breadth, HPO terms}) = -0.007$ **B****The rule is skeletal, not general**

Breadth of GWAS action



Breadth of the Mendelian phenotype

C**A shared layer needs coherent sites**

103 genes act at every site

no gene acts at every volume